10.1 Introduction of Automation

Automation is the technology by which a process or procedure is performed without human assistance.

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Automation covers applications ranging from a household thermostat controlling a heater, to a large industrial control system.

Automation has been achieved by various means including mechanical, hydraulic, pneumatic, electrical, electronic devices and computers, usually in combination. Complicated systems, such as modern factories, airplanes and ships typically use all these combined techniques.

The word "Automation" is derived from ancient Greek words of Auto (means "Self") Matos (means "moving"). Thus, a mechanism move by itself or self dictated is called automation.

10.2 Types of Automation

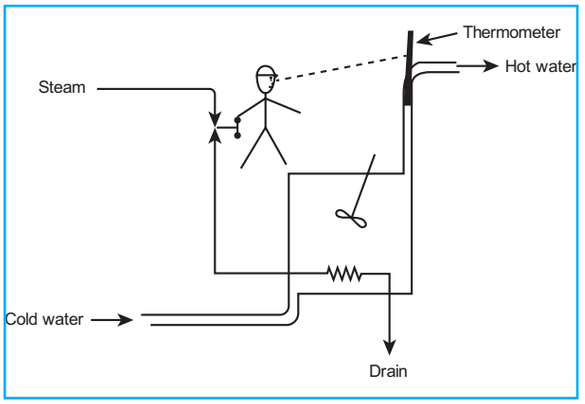
1. **Based on control**

a. Partial Automation

It means replacement of human activities or involvement by automatic means only partially.

b. Fully automation

The human involvement is totally eliminated and the process is entirely carried out and controlled through automatic means along with a proper feedback system.



Partial Automation

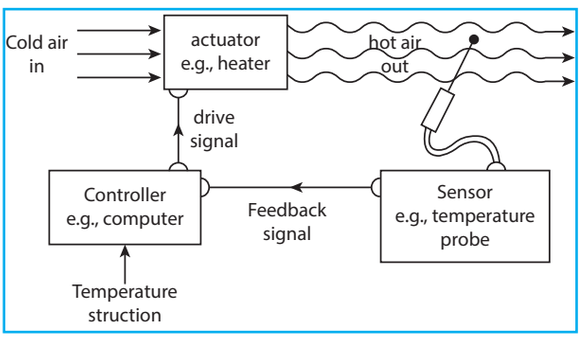
2. **Based on Application**

a. Building Automation System (BAS)

It is the automatic centralized control of building's heating ventilation, air conditioning and lighting and other systems through a building automation system (BAS). The classifications are

i. Home Automation

ii. Office Automation



Whole Automation

b. Industrial Automation

It is use of computers and robots to control industrial processes like manufacturing without significant human assistance. The classifications are

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